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II. RESEARCH METHOD

A. Photoplethysmograph

Photoplethysmograph (PPG) is a method to determine blood volume from skin tissue. The PPG tool depicts in Fig. 1.

In Fig. 1, The PPG application uses an emitter (light source) or LED on the module by utilizing the back wave from the reflection of human skin to the photodiode receiver (light sensor) [13]. This method uses an optical sensor to capture electrical signals from reflected light sources due to changes in blood flow during the heart work [14]. The sensor obtained peak waves with different maximum heights depending on the blood flow volume.

On a human finger, there is blood flowing. The blood will be reflected by light and then captured by light-sensitive sensors. Blood concentration, flow, and blood pulse are factors that affect the high and low of the sensor readings.

B. Random Forest

Random forest (RF) is an algorithm for predicting data based on decision trees. RF prediction is carried out by merging trees (trees) and conducting training on their sample data [15]. Using more trees will affect the accuracy obtained for the better. The illustration of RF is depicted in Fig. 2.

In Fig. 2, Decision-making with Random forest is taken based on the voting results of each decision tree. The decision results are taken from the voting of each tree. Random forest can set the number of trees formed to get the best accuracy with several evaluations [16][17]. A tree is automatically formed with data distribution with the same class. The process of determining the decision is done after the tree is formed. After having decisions from all trees, initialization is carried out on all decisions still for each tree by adding and dividing by the number of trees formed.

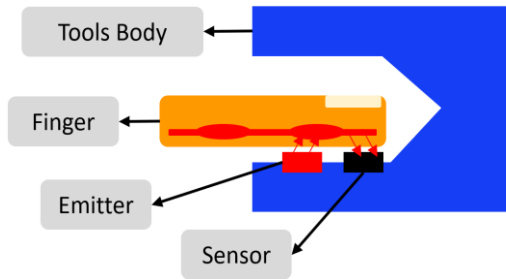


Fig. 1. PPG Tool

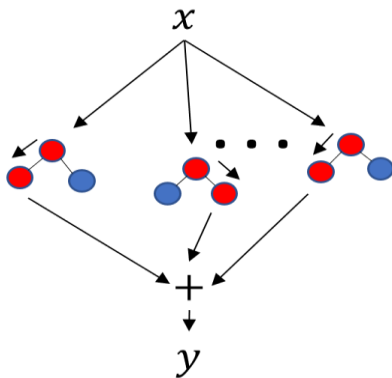


Fig. 2. Illustration of Random Forest

$$rf = \frac{\sum_{i \in all\ tress} T}{T} \quad (1)$$

where:

rf : random forest output

T : total used trees

$all\ tress$: total output tree

Like equation (1), the total of the vote results will be added up and divided by the total prediction tree. The results of each decision are obtained from the formed output tree. The number of trees can affect the accuracy of the results. The more trees, the more decisions that can improve accuracy but with a note that the calculation process is also heavier. It is necessary to experiment several times to get the best performance in the calculation process and the model.

C. Moving average Filter (MAF)

This filter is the most common in the case of Digital Signal Processing (DSP) because of its easy implementation. MAF is a method of trimming and imputing in fixing values. Despite its simplicity, this MAF is optimal for reducing random noise while maintaining a sharp step response. MAF is depicted in Fig. 3.

In Fig. 3, Blue is the original signal, and red is the MAF result. Noise in the signal can be reduced using MAF. This makes the MAF the main filter in the case of signals with the time domain [18]. The application of MAF has good performance in processing because it has simple calculations. In this case, a MAF is used to reduce the initial noise so that significant spikes do not affect the features acquired.

In the application of MAF, the thing that needs to be considered is the frequency of the data taken to be divided because the more or closer the data is divided by the following data, the finer the data will be. Clean and refined data can be a positive thing, and it can be a negative thing. It becomes positive because the data is clean from noise and becomes negative because it could be that the smoothed data is too clean, there are missing features, or important information is missing in the signal.

D. Slope

The Slope shows the size of the angle formed from the difference in elevation of a graph, which is usually expressed in percentages or degrees. Slope is depicted in Fig. 3.

Fig. 3, illustrates how the Slope is represented as the ratio of the vertical change between two points to the horizontal change between the two points. Slope can overcome outliers caused by the Slope of the data. Outliers due to the Slope of the data will not be used because the data does not have characteristics that describe the correct value in the sample.

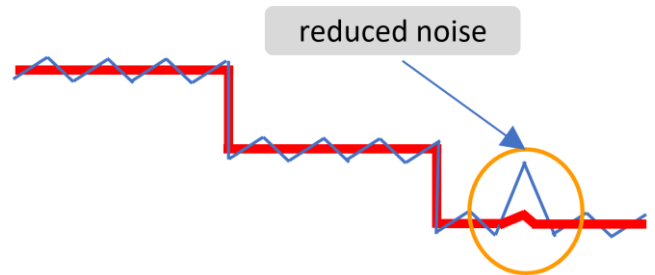


Fig. 3. Illustration of Moving Average Filter (MAF)

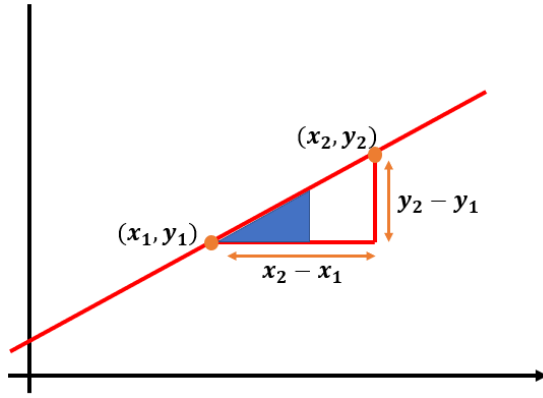


Fig. 4. Slope

The slope value is obtained using equation (2). The Slope (m) is expressed by the current y value minus the previous y value and divided by the current x value minus the previous x value.

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad (2)$$

where:

- m : slope
- y : point y on the graph and number one for start, number two for end point
- x : point x on the graph and number one for start, number two for end point

This study uses the Slope to see whether a slope appears in the existing data. This Slope indicates a spike in the increase in sensor data values. The PPG signal has the form of a pulse so that a spike will appear up to a certain point, but if the spike that appears is very high, it will make bad data.

E. Z-score

Z-score is a calculation method that describes the relationship of a value with the average of a group of values. The Z-score is measured in terms of standard deviations from the mean. This study used a 95% confidence level, meaning that valid data must be in the Z-score range of -1,960 to 1,960.

$$Z = \frac{(x - \mu)}{\sigma} \quad (3)$$

where:

- Z : Z-score
- x : observed value
- μ : mean
- σ : standard deviation

Determining the value of the Z-Score can use equation (3) where the Z-score is the difference between a value and the average value of a population divided by the population's standard deviation. The variation of the data influences the high and low divergence. The more the data varies, the higher the deviation [19].

F. MAPE

Mean Absolute Percentage Error (MAPE) is the absolute (absolute) average percentage error. The definition of Mean Absolute Percentage Error is a statistical measurement of the accuracy of estimates (predictions) in forecasting or regression methods. This evaluation method is straightforward and can be used to determine the performance of machine learning regression in the form of the percentage error obtained from the prediction results with the actual results [20].

$$MP = \frac{1}{n} \sum_{t=1}^n \left| \frac{y_i - \tilde{y}_i}{y_i} \right| 100\% \quad (4)$$

where:

- MP : mean absolute percentage error
- n : total data
- y : actual value
- $\hat{y}$: predict value

As shown in equation (4), MAPE can determine the error value based on the difference between each actual and predicted value, then add up and divide by the entire data. The use of MAPE makes it easier to evaluate by looking at the percentage of error of each learning used, especially regression, because it has an output in the form of a value that is close to or equal to the training value. From the MAPE value, an error value will be obtained in the form of a percentage. For the result to obtain an accuracy value, the error value is subtracted by 100%.

III. PROPOSED METHOD

The proposed method in this research is to determine outliers as valid data and invalid data is depicted in Fig. 5.

A. Outlier detection

Outlier detection consists of three filters applied. The first filter is to detect noise in the signal. Little noise will be handled by the MAF, while high noise considered an outlier; an example of noise data is depicted in Fig. 6 (a). The second filter is to see the Slope of the signal using the Slope detection method; an example of skewed data is depicted in Fig. 6 (b). Furthermore, the third is to use the Z-score to detect unreasonable signals, such as being far from the distribution of other values; an example of the distribution of unreasonable values can be seen in Fig. 6 (c).

Based on Fig. 6. outliers can be of various types, and this cannot be done if you only use one method. If outlier detection only uses an abnormal filter, then outliers in the form of ripples from the data may not be visible because each data has hundreds of values in one output, so you have to use more than one method to eliminate outliers.

In TABLE I, the pseudocode of the filter explains the several logical stages used in determining filter 1, filter 2, and filter 3.

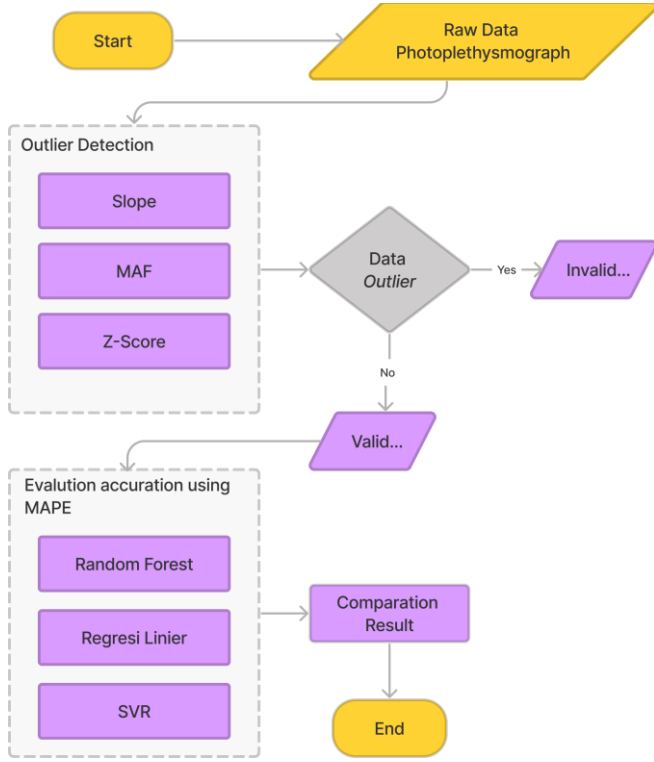
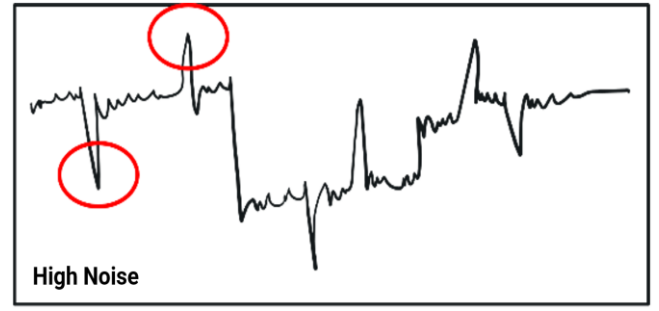


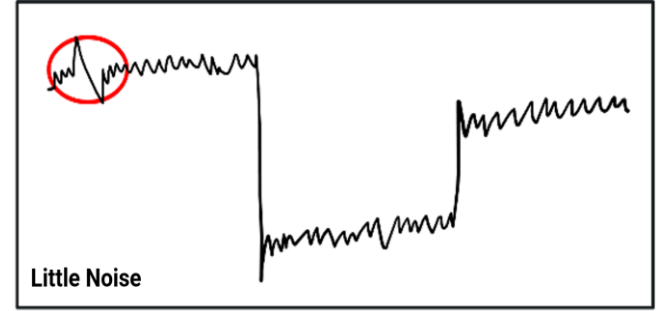
Fig. 5. Proposed Method

TABLE I. PSEUDOCODE OF FILTER

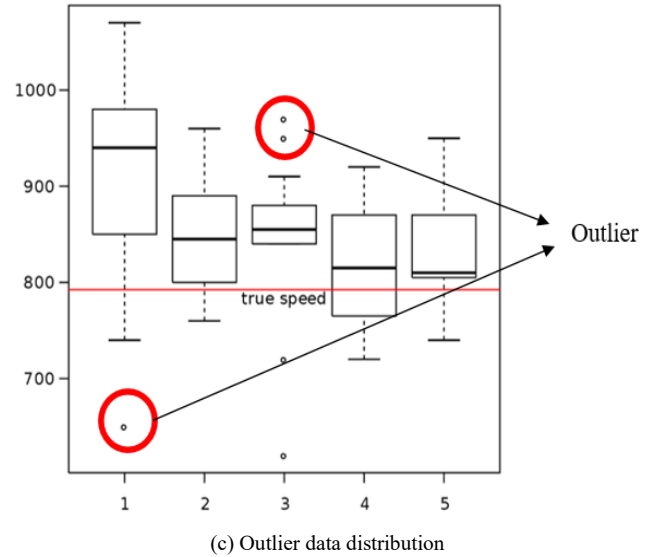
Pseudocode of Filter
<pre> readSensor ← read sensor data form PPG device total ← total add readIndex readIndex ← readIndex add 1 increment ← for increment index data for totaldata for range(200) ← for handling Red sensor 0-199 Filter 1() ← function filter using MAF if readIndex >= readIndex+numReading ← data read read index=0 ← check number data maf=sigma n(xn-xn-1)/n ← count of MAF endif Filter 2 () ← function filter using Slope if (n>10) ← if the data n is more than ten, count Slope slope=abs((dataRead[n]-dataRead[n-10])/10) endif if(slope>8) ← if the number of the slope is more than eight, the data is passed, and if less than eight, the data is invalid endif increment endfor for range(200) ← for handling Green sensor 200-399 Filter 1() Filter 2() endfor for range(200) ← for handling Blue sensor 400-599 Filter 1() Filter 2() endfor endfor calculationZscore() ← for calculation Z-score </pre>



(a) High noise



(b) Little noise



(c) Outlier data distribution

Fig. 6. (a) High noise, (b) Little noise, (c) Outlier data distribution

IV. EXPERIMENTAL RESULT

A. Material Data

Material data is photoplethysmograph from sampling on the PPG tool with 1248 data. For each data, there are three sensor output signals. Each signal has 200 data samples in R, G, and B data. For details datasets are depicted in TABLE II.

In TABLE II, R, G, and B output sensor will be processed into features in outliers detection. A line graph will form 200 data for each output to find out noise, abnormal data and others to make sure the initial stages of the data are bad or good for training data machine learning using Random Forest.

TABLE II. DATASET PHOTOPLETHYSMOGRAPHY FROM SENSOR

Number	Output Sensor (Milivolt)		
	Red	Green	Blue
1	2467	514	1878
2	2491	523	1878
3	2516	515	1878
...	...	...	...
200	2564	525	1874

TABLE III. RESULT OF FILTER

Tools	Dataset	Two Filter	Three Filter
Dataset A	571	462	372
Dataset B	677	640	386
Total	1248	1102	758

TABLE IV. COMPARISON RESULT ACCURACY

Method	No Filter	Two Filter	Three Filter
Linear Regretion	71.54%	78.31%	81.20%
Support Vector Machine	73.44%	80.89%	85.54%
Random Forest	75.43%	89.65%	93.99%

Each data collection has 600 data consisting of 200 red values, 200 green values and 200 blue values. The collection was carried out in two places, separated into dataset a and dataset b, shown in TABLE III. In TABLE III, shows different values when filtering the outlier data. The data results before and after being filtered are processed into machine learning. The comparative results of each machine learning are shown in TABLE IV.

In TABLE III shows the results of outlier detection with first filter, second filters, and third filters. The total of all data is decreased, which indicates that there is a reduction in data due to the outlier filter.

The data obtained will be less than the overall data because each outlier filter will reduce the data. At the beginning of data collection, there is usually a reduction in value that is more than the next, which is caused by several factors, namely inappropriate sampling, inappropriate finger positions, or other problems.

B. Evaluation

The evaluation using MAPE is applied to show the error of the proposed method can improve accuracy in Linear Regression, Support Vector Machine, and Random Forest. Input for machine learning is the output result of all data and the output result of valid data after removing outliers. The evaluation process in this study was carried out by comparing the results of each model's accuracy. In table TABLE IV, several methods have been done before the outlier filter is carried out. Linear regression and SVR have a model performance of accuracy get less than the Random forest method for the data without the filter. Random forest without a filter has a performance of 75%, better than the previous method, so this method was tested in the outlier filter experiment. The random forest method has the best performance, with an accuracy of 93.99% with three filters in the proposed method.

V. CONCLUSIONS

Outlier detection with the proposed combined method successfully determined valid and invalid data (outliers). from a total of 1248 data in the dataset used to make outlier detection. Applying the MAF method to filter small and large noise data, then the Slope method to detect skewed

data, and the Z-score method to determine the outliers of unreasonable data against other data distributions. Comparison results show improved results for each filtration process, with the best results using a combination of the three methods. The number of outliers detected was quite prominent in sampling from the PPG tool based on the experimental results. Therefore, with the proposed method, outlier detection can be used for early warning in sampling so that incoming data can be accepted and rejected automatically in a device to reduce values discarded due to outliers.

In future work, we will implement an invalid or outlier detection signal so that the tool can determine invalid data to reduce data outliers so that the incoming value has good data quality.

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